Exam Date & Time: 13-Aug-2022 (10:00 AM - 01:00 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. EXAMINATIONS (MIT MANIPAL) - AUGUST 2022
SUBJECT : BIO 1051 - BIOLOGY FOR ENGINEERS
(MAKEUP)

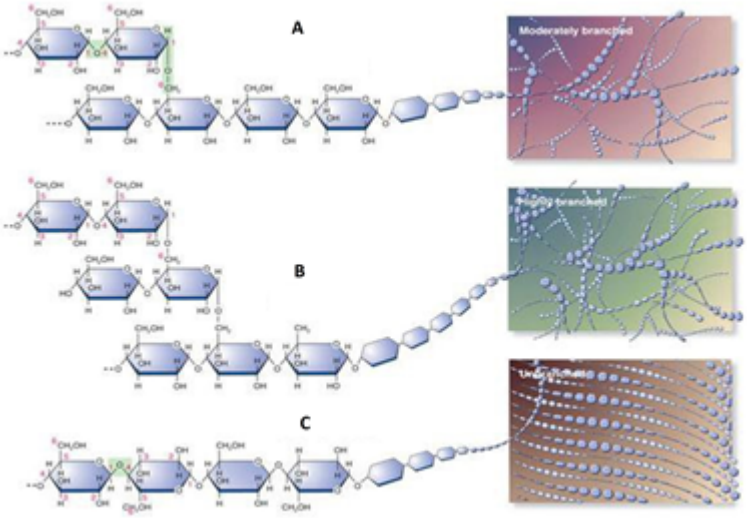
BIOLOGY FOR ENGINEERS [BIO 1051]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

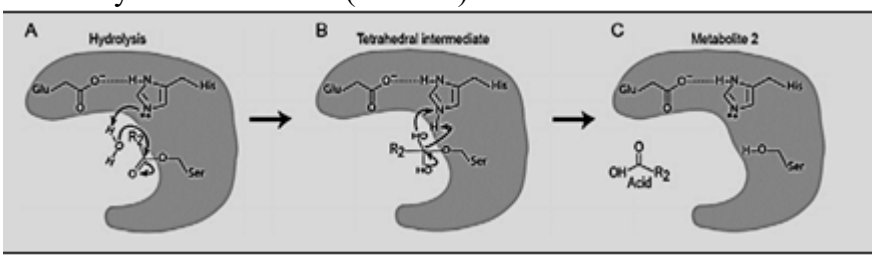
1A)



(3)

Given above are the chemical representation of three types of molecules.
(1) What is the purpose of compound A and B for living things ? (1 mark)
(2) What makes the difference in these compounds? (1 mark)
(3) Will you agree with the statement "Compound C can form inter chain interactions". Justify your reasoning (1 mark)

1B) Depicted below is the enzyme substrate reaction. Carboxylesters are cleaved by carboxylesterases via the addition of water to form an alcohol and an acid metabolite. The image represents the concluding steps in the hydrolysis reaction of a hypothetical carboxylester, leading to the formation of an acid metabolite. Observe the image and answer the following questions:
i) If we change the enzyme with a different enzyme, what will happen to the reaction? Justify (1 mark)
ii) If histidine is substituted by (a) lysine or (b) phenylalanine, which would reduce the efficiency of the reaction? (2 marks) (3)

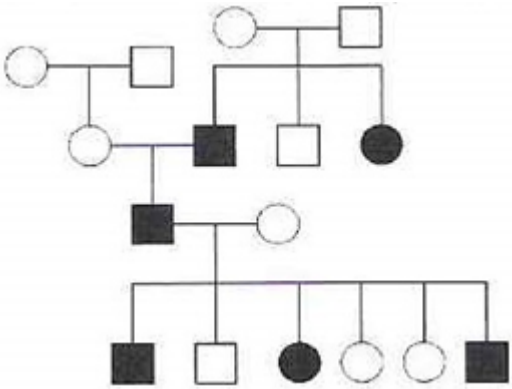


1C) Fructose 6 phosphate→ Glucose 6 phosphate (4)
i) Calculate ΔGo' for the reaction at 25°C.

ii) Concentration of fructose 6 phosphate is adjusted to 1.5M and glucose 6 phosphate to 0.5 M. Now calculate ΔG?
iii) According to you ΔGo' and ΔG are same or different? Justify your reasoning.

2A) In pea plants round seed shape is dominant to the wrinkled seed shape and yellow seed color is dominant to green seed color.
i) What will be the phenotype(s) of F1 progeny if you cross true-breeding plants showing Round, Green seeds with a true-breeding Wrinkled, Yellow seeds? (3)
ii) You crossed the F1 progeny to give F2 progeny as per the Mendelian model. Show this cross using a Punnett square.

2B)



(3)

Analyse the above mode of inheritance and answer the following questions
(i) Why affected children have both unaffected parents? 1 mark
(ii) What is the reason that both male and female are affected ? 1 mark
(iii) What is the most probable mode of inheritance? 1 mark

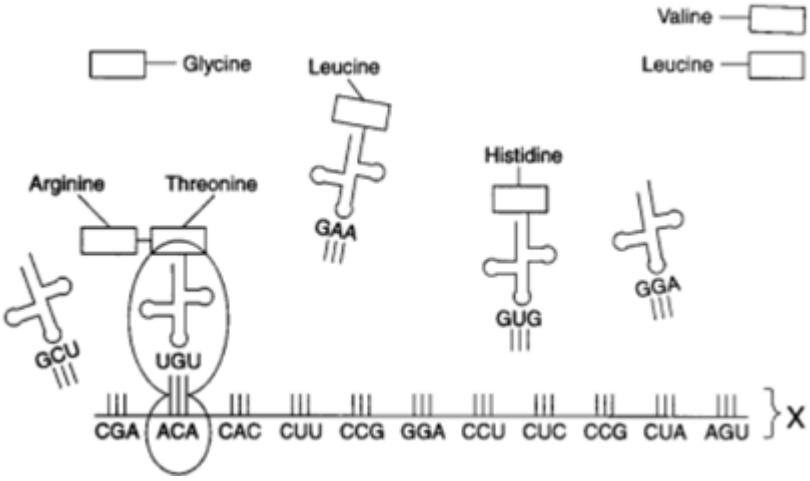
2C) In a new species of fruit fly the curly wings phenotype is a mutant and normal wings is wild type. Assume that the study is for flies that are true breeding for the curly wing phenotype and flies are true breeding for normal wings
i) What phenotype is expected to be seen in the F1 generation if the mutant phenotype is dominant? In what proportion would you see these phenotypes?
ii) What phenotype(s) would you expect to see in the F1 generation if the mutant phenotype is recessive? In what proportion would you see these phenotypes? (4)
iii) A homozygous purple flowered short stemmed plant was crossed with a homozygous long stemmed red flower color plant and the F1 phenotypes had purple color with short stem. This F1 generation was crossed with double homozygous recessive plant. Observe following progeny and explain the results for association between the two characters. Purple flower short stemmed- 52; Purple flower long stem- 47; Red flower short stem- 49; Red flower long stem- 45.

3A) In Hershey and Chase experiment, what kind of organisms were used? What is the reason for selecting ^{32}P and ^{35}S labelling (in) for DNA and protein respectively? A researcher revisited the same experiment with the isotopes of carbon and oxygen. What would be the outcome of this study? (3)

3B) i) Based on the structure and/or chemistry of DNA or RNA give two reasons for why DNA is a better choice to be the stable, heritable genetic material than RNA. (2 marks) (3)
ii) If we heat a DNA molecule which bond will break first? (1 mark)

3C) Using the below given figure, answer the questions: (4)
i) Where the structure "X" is made?
ii) In which cell organelle the process represented in the diagram occurs?
iii) Which amino acid will be transferred to position of codon "CAC"

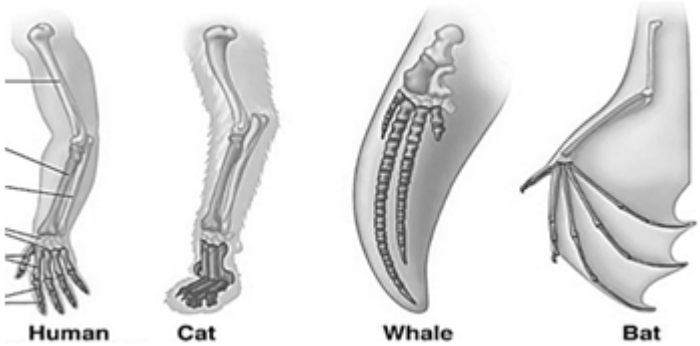
iv) What kind of codon change occurs due to a silent mutation at codon "CAC"



4A) Analyze the following molecules. Will you agree with the statement "Living systems achieved different functions by a simple chemical modification of a basic structure". Justify (3)

i) Chlorophyll
ii) Hemocyanin
iii) Myoglobin

4B) The figure depicts skeletal structure of forelimbs from different animals. (3)



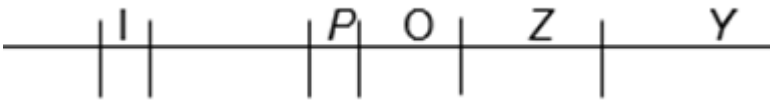
i) Why is it that the skeletal structure is so drastically different in these animals, even though the skeletal structure indicates that they have evolved from a common ancestor?
ii) Can these forms be categorized as incremental changes? Comment on the form and function of these skeletal structures.

4C) What bioinspiration can be made from the following which can be applied to various areas of engineering? (4)
a) Woodpecker b) Touch me not plant; c) Chitin (compound in exoskeleton of some animals); d) Snake

5A) As a boy suffered from a viral disease. 5 years later, he was again sick from the same viral disease. The first time he had the infection, it was there for 10 days. But the encounter lasted for only 2 days with mild symptoms. What could be the reason for this improved immunity? What type of immunity is this? (3)

5B) What will be the effect on the plant if the xylem vessels were composed of living cells? (3)

5C) Following is an illustration found in the lecture note book of a "Biology for Engineers" student. (4)



i) What does the illustration represent? Explain (2 marks)

ii) Is there any importance in the correct order of different parts such as POZY? (1 mark)
iii) If we delete the part "P" whether the system will work? Justify (1 mark)

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